



IILM UNIVERSITY, GREATER NOIDA

INTERNSHIP REPORT PRESENTATION

Deloitte Technology & Forensic Analytics

Virtual Internship

Deloitte Australia Technology Virtual Internship Program | Hosted on Forage

YASHASHWAI

Roll No. 2410031395 | B.Tech CSE

July – August 2026



YASHASHWAI

B.Tech – Computer Science & Engineering

ABOUT THE STUDENT

Candidate Profile



Roll Number

2410031395



Institute

IILM University, Greater Noida, U.P.



Programme

B.Tech CSE, Batch 2024–2028



Internship Domain

Deloitte Technology & Forensic Analytics



Duration

9 Weeks | July – August 2026

ABOUT THE PROGRAM

Deloitte, Forage & Daikibo Industrials

Deloitte is one of the world's leading professional services networks, delivering audit, consulting, financial advisory, risk advisory, and tax services worldwide. This virtual internship was hosted on Forage, an open-access platform that partners with global employers to build practical, self-paced work-experience simulations centered on Daikibo Industrials — a fictional international manufacturing client operating automated plants across Tokyo, Osaka, Berlin, and Shenzhen.



Deloitte

Global professional services network specializing in cutting-edge analytics and forensic audit frameworks.



Forage

Open-access virtual work-experience platform delivering industry-aligned, self-paced simulations.



Daikibo Industrials

International manufacturing client with automated plants in Tokyo, Osaka, Berlin & Shenzhen.

Turning Raw Telemetry Into Operational Intelligence

Modern smart manufacturing facilities deploy thousands of automated IoT sensors across shop-floor machinery. Telemetry is stored in raw, multi-level JSON structures that make it difficult to extract operational intelligence, diagnose downtime bottlenecks, and perform root-cause analysis — while rising internal complaints called for a data-driven review of gender pay equity across global job roles.

This internship addressed that gap — turning raw JSON telemetry into an interactive dashboard and a forensic pay-equity model:



What the Internship Set Out to Achieve



Ingest & Validate Telemetry

Parse and validate 160,704 JSON sensor records across 9 machine types in 4 global factories.



Engineer the Downtime Measure

Formulate a Tableau calculated field that translates health-status messages into downtime minutes.



Build Downtime Visuals

Construct bar charts comparing downtime at the facility level and the device level.



Integrate an Interactive Dashboard

Combine the charts into a Tableau dashboard with dynamic cross-filtering capability.



Use-as-Filter Actions

Connect Sheet 1 to Sheet 2 so factory selection filters the device-level breakdown.



Classify Pay Equity

Analyze compensation across 38 job roles and classify Fair, Unfair & Highly Discriminative tiers.

The Toolkit Behind the Internship



Tableau

Dashboard & BI visualization



Microsoft Excel

Spreadsheet logic modeling



Python 3.12

Data verification scripting



openpyxl

Excel file automation



pandas

Data wrangling & analysis



matplotlib

Verification plotting



JSON / XLSX

Telemetry & role data formats



HTML5 · CSS3 · JS

Local dashboard simulation

From Raw Telemetry to Forensic Decisions



Data Ingestion Layer

daikibo-telemetry-data.json (160,704 records) & Task 5 Equality Table.xlsx (38 role profiles)



Transformation & Measure Layer

Tableau measure IF [Status] = 'unhealthy' THEN 10 ELSE 0 END, plus pay classification logic



Visualization & BI Layer

Factory & device-type sheets unified into a dashboard via a “Use as Filter” action



Decision & Forensics Layer

Identified Daikibo Seiko & the LaserWelder as top drivers; flagged 7 discriminative roles

The 6-Module Project Workflow

1

MODULE

Data Ingestion

Imported multi-level JSON telemetry, checking all schema levels.

2

MODULE

Measure Engineering

Created the Unhealthy field: 10 minutes per unhealthy message.

3

MODULE

Factory Sheet Design

Horizontal bar chart of cumulative factory downtime, sorted descending.

4

MODULE

Device Sheet Design

Horizontal bar chart of device-type downtime, sorted descending.

5

MODULE

Dashboard & Filter Action

Combined sheets on a dashboard with a Use as Filter action.

6

MODULE

Pay Equity Modeling

Classified 38 job roles by statutory pay equality score brackets.

What the Analysis Revealed

1,030 min

Total downtime across all four factories in May 2021
(17.2 hrs)

daikibo-seiko

Highest-downtime plant — 480 min, 100% caused by
the LaserWelder

37 roles

Compensation profiles classified: 19 Fair · 11 Unfair · 7
Highly Discriminative



Practical ability to translate raw multi-level JSON telemetry into Tableau calculated fields



Hands-on experience building cross-filtering, factory-to-device drill-down dashboards



Skill in applying nested IF/ABS logic in Excel to build a defensible pay-equity classifier



Capability to identify root-cause drivers of operational downtime from sensor data

CERTIFICATION

Internship Successfully Completed



Tasks Completed

Data Analysis & Forensic Technology



Enrolment Verification

6a943cb62eef4b8fc0346b05



User Verification Code

6a943b5c2eef4b8fc0346ce9



Duration

July 1 – August 30, 2026

Deloitte.

Yashashwai Chaudhary Data Analytics Virtual Internship

Certificate of Completion
From July 1 to August 30th, 2026

Over the period of July and August 2026, Yashashwai Chaudhary has completed practical tasks in:

Data analysis
Forensic technology

Tina McCreery
Chief Human
Resources Officer,
Deloitte

Enrolment Verification Code 6a943cb62eef4b8fc0346b05 | User Verification Code 6a943b5c2eef4b8fc0346ce9 | Issued by **Forge**



Thank You

Deloitte Technology & Forensic Analytics Virtual Internship | Hosted on Forage

YASHASHWAI

Roll No. 2410031395 | IILM University, Greater Noida